Low Dropout Regulator

Electronics project report presented to Eng. Ahmed Amr

Team 8

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1. Introduction

This report presents the design, hand analysis, and simulation verification of a **low-power two-stage OTA** for use in a low-dropout (LDO) regulator. The OTA is implemented in **65 nm CMOS technology** and operates from a **1.2 V supply**, meeting the target specifications for gain, bandwidth, stability, input common-mode range, CMRR, and power consumption.

.2. Design Specifications

2.1 Technology Parameters

- Technology: **65 nm CMOS**
- Simulator: **Cadence Spectre**
- Supply Voltage (VDD): **1.2 V**
- Process Corners: **TT, SS, FF**

3. OTA Architecture

3.1 Topology Selection

A two-stage OTA topology was selected for this design. This architecture provides sufficiently high DC gain while remaining compatible with low-voltage operation at a 1.2 V supply. In addition, the two-stage structure allows the use of Miller compensation, which is well suited for stabilizing the amplifier when driving capacitive loads. The chosen topology also enables low power consumption, making it appropriate for LDO applications.

3.2 Circuit Description

The OTA consists of a differential input stage that converts the input voltage difference into a transconductance. An active current mirror load is used in the first stage to increase gain. The output of the first stage drives a second gain stage, which further amplifies the signal and drives the load. To ensure stability in the presence of capacitive loading, a Miller compensation capacitor is connected between the output of the second stage and the intermediate node.

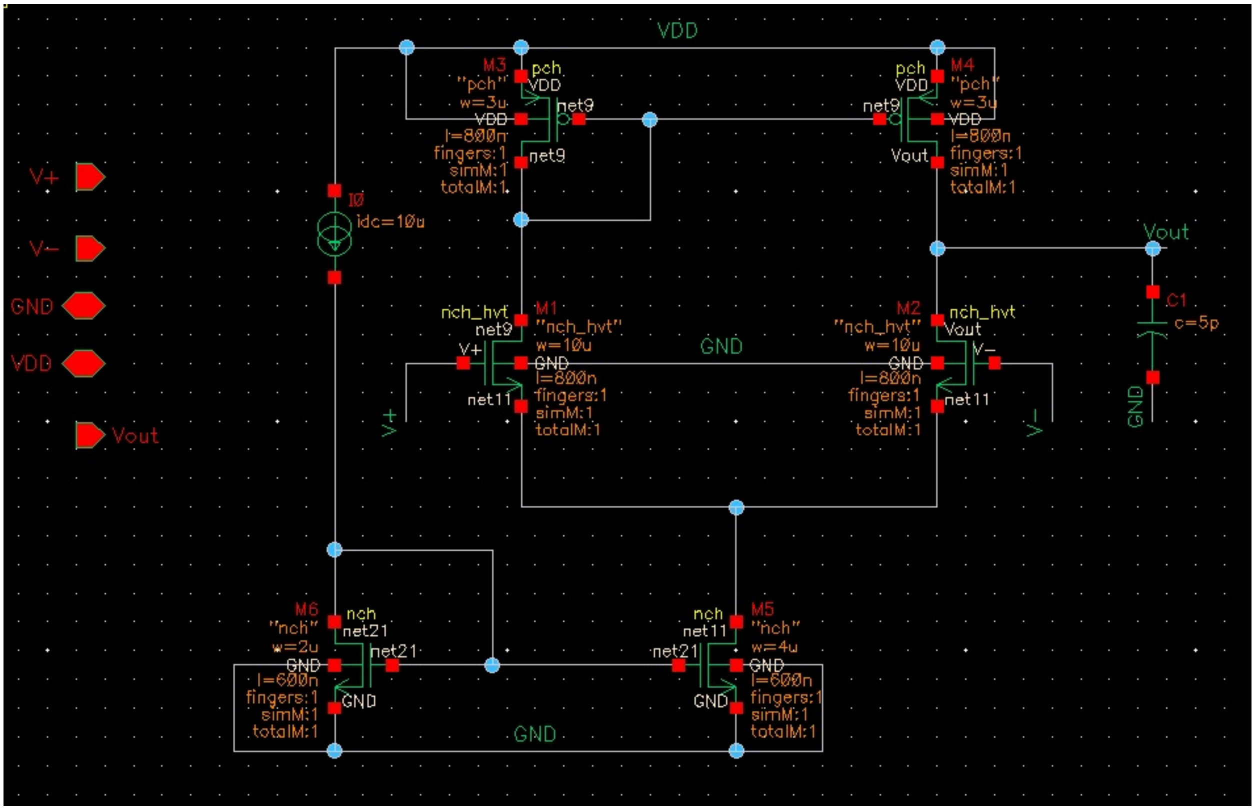


Figure 1: OTA Schematic

4. Design Methodology

4.1 Biasing Strategy

A 10 µA reference current generates all OTA bias currents using current mirrors. The total OTA current is kept below 60 µA. The tail current source is biased at gm/ID = 13, while the current mirror loads operate within a gm/ID range of 8 to 15 to ensure robust operation across process corners.

4.2 Device Sizing Approach

A gm/ID-based design methodology is used to systematically meet gain, bandwidth, and power specifications.

Input Differential Pair (M1–M2)

we used nch_hvt (NMOS high Vth) to easily achieve the dc gain as it has higher Vearly with respect to nch (NMOS standard Vth).

- $I_{D1,2} = 10\mu A$

- $GBW = \frac{g_{m1,2}}{2\pi C_L}$
- $g_{m1,2} = 2 \cdot \pi \cdot 5p \cdot 5\text{MHz} \approx 0.16\mu S$
- $\frac{g_m}{I_D} \approx 16\ S/A$

To identify **L** we assume **gds** for **NMOS** and **PMOS** are equal

- $A_v = \frac{g_m r_o}{2} = \frac{g_{m1,2}}{2 g_{ds1,2}} \geq 50$
- $\frac{g_{m1,2}}{g_{ds1,2}} \geq 100\ ,\ g_{ds1,2} \leq 1.6\mu S$

from **gm/gds vs gm/id** chart we got **L =** $\mathbf{0.8\mu}$**.**

from ID/W vs gm/id chart we got W = 10µm.

Current Mirror Load (M3–M4)

The PMOS loads design by assuming $\frac{g_m}{I_D}$ = 15 S/A

- $g_{ds3,4} = \frac{I_D}{V_A} \leq 1.6\mu S\ \ ,\ V_A \geq 6.25$

from **gm/ID vs gm/gds Pmos** chart we got **L** = $\mathbf{0.8\mu}$

- $CMIR_{high} = VDD - V_{SG3,4} + V_{thn} \geq 1$
- $V_{sg3,4} \leq 1.2 - 1 + 0.472\ ,\ V_{sg3,4} \leq 0.67$

From **(gm/ID)** vs **VGS** we got **(gm/ID)min = 8 S/A** then from **gm/ID** vs **ID/W**

We got **W =** $\mathbf{3\mu}$**m**

Tail Current Source (M5)

For **ISS** = $\mathbf{20\mu A}$ and assuming **gm/ID = 13** for moderate inversion.

- $CMRR = \frac{A_v}{A_{vCM}} = 45dB$
- $A_{vCM} \approx \frac{-1}{2 g_{m3,4} RSS} = -11dB$
- $g_{ds5} = 45\mu S\ \ ,\ \ \frac{g_m}{g_{ds}} \geq 7$

Then L = **0.6**$\mathbf{\mu}$**m**

- $CMIR_{low} = V_{gs1} + V_{dsat5} \leq 0.65\ V$
- $V_{dsat5} \leq 0.125\ V$

We got **(gm/ID)min = 11.4**, so we chose **(gm/ID) = 13** to keep some margin.

From ID/W design charts we got the required **W = 4**$\mathbf{\mu}$**m**.

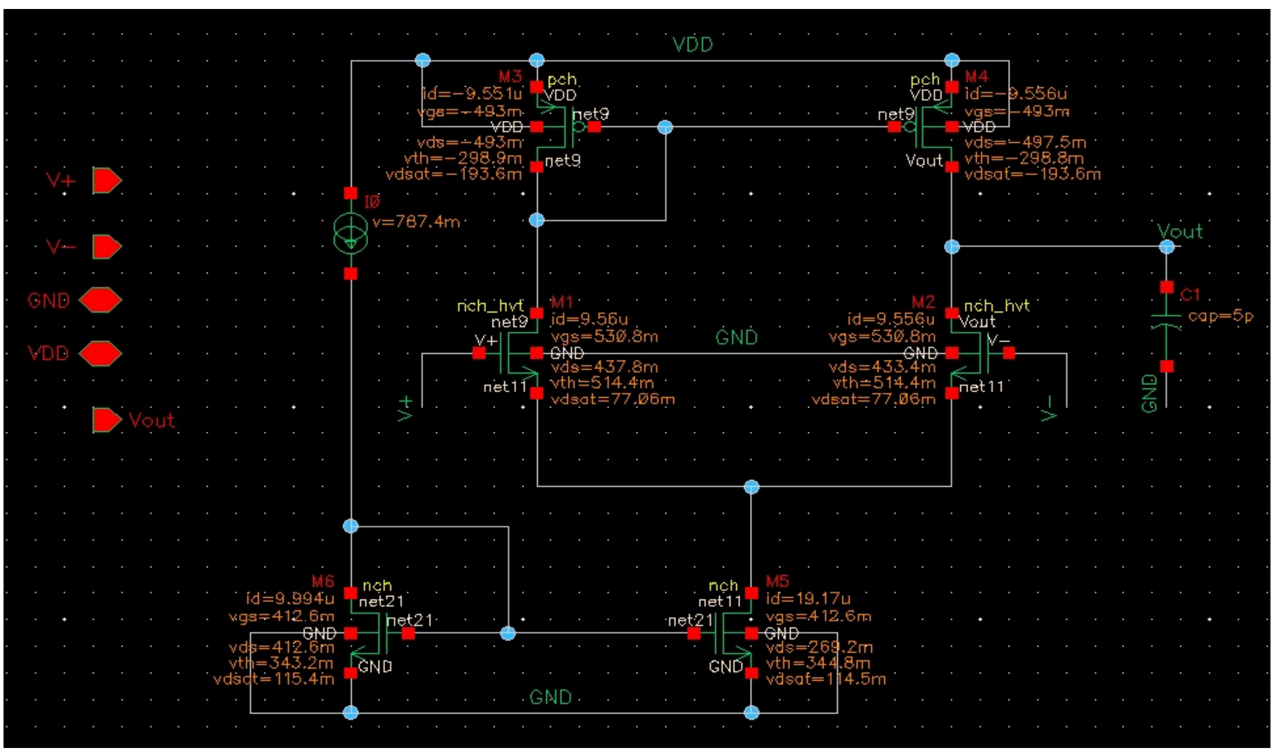
This ensures sufficient **CMRR ≥ 45 dB** while maintaining the required input common-mode range.

- **Verification**
  Final device dimensions are validated through DC operating-point simulations to confirm saturation and compliance with all specifications

Transistor	Input Pair	PMOS load	Tail CS
L(μm)	0.8	0.8	0.6
W(μm)	10	3	4
ID(μA)	10	10	20
gm/ID(S/A)	16	15	13
Vdsat(mV)	77	190	115

5. Simulation Results of the OTA

5.1 DC Operating Point



<u>Figure 1: DC Operating Point & MOSFETs' currents</u>

5.2 DC Gain Analysis

DC gain = 38.5 dB

GBW = 5.57 MHz

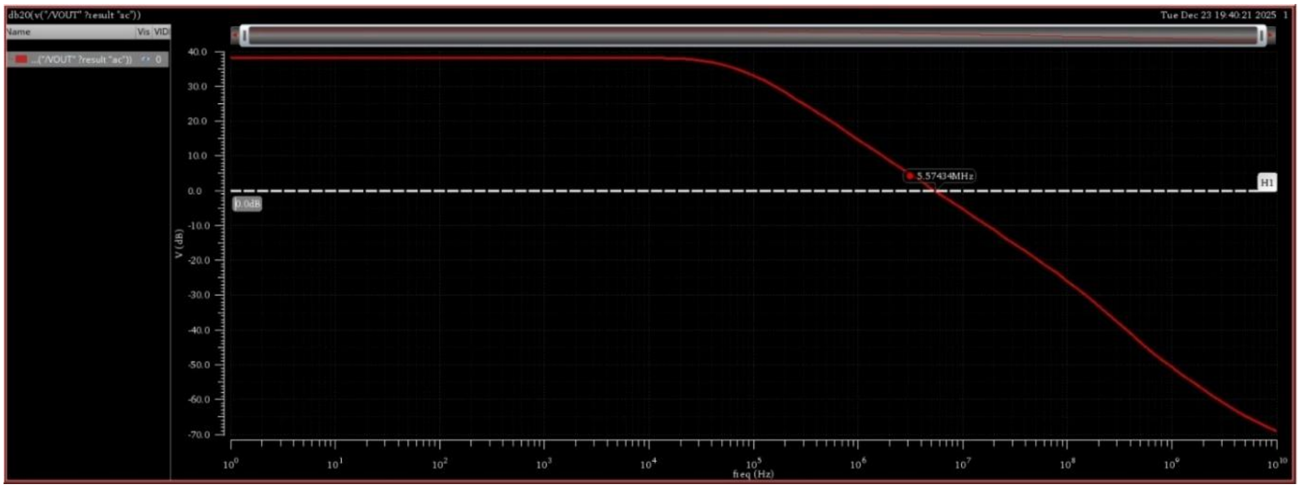


Figure 2: DC gain

5.3 Common-Mode Input Range (CMIR)

$CMIR_{low} = 0.62$

$CMIR_{high} = 1.2$

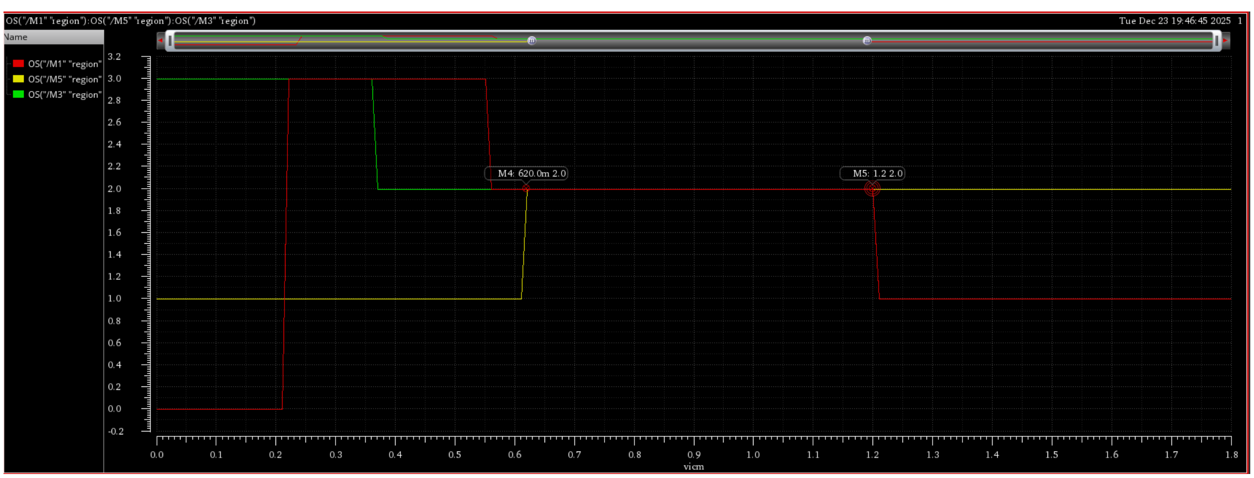
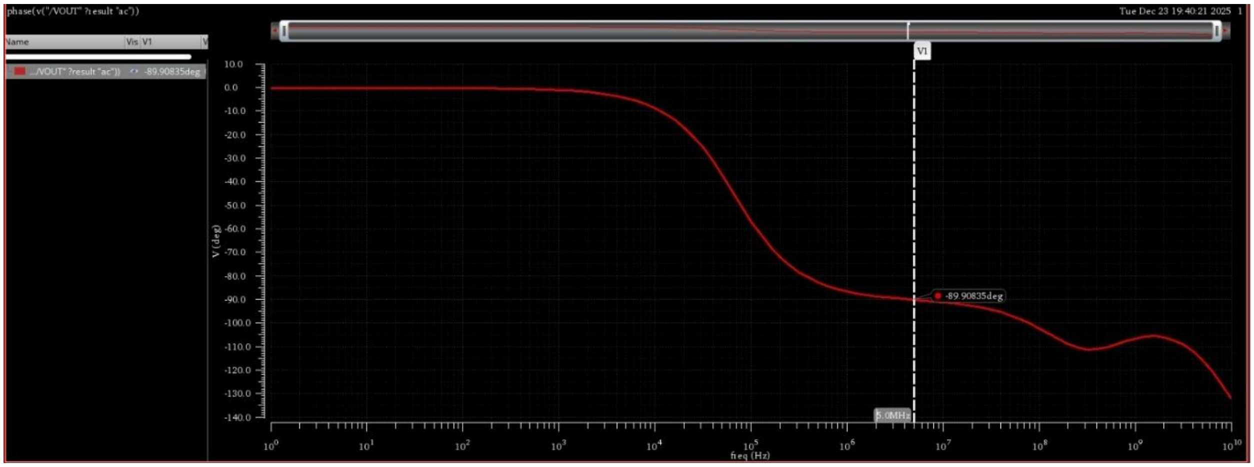


Figure 3: CMIR

5.4 Stability and Phase Margin

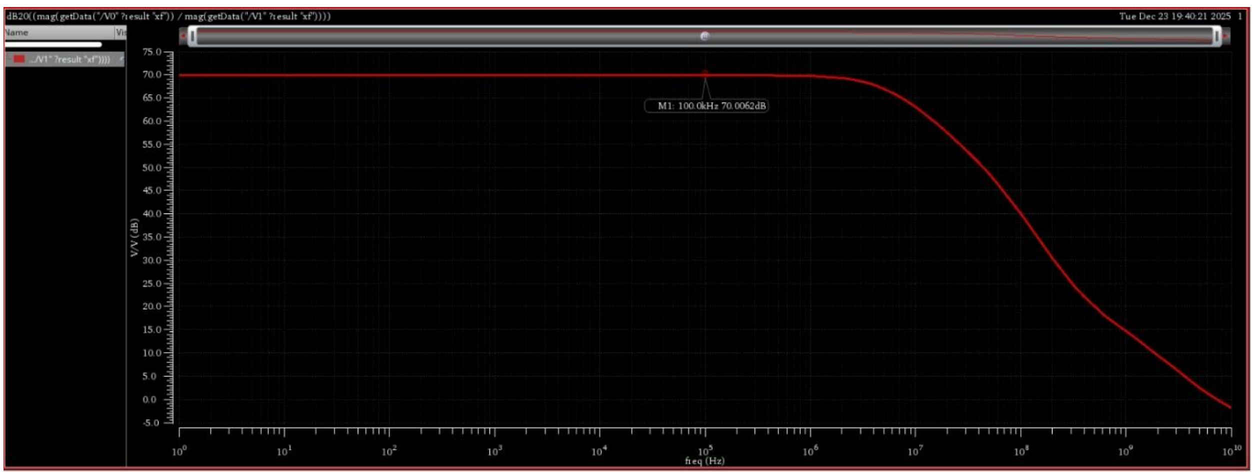
Phase Margin = 89.9 deg



<u>Figure 4: Phase Margin</u>

5.6 Common-Mode Rejection Ratio (CMRR)

CMRR = 70 dB



<u>Figure 5: CMRR</u>

5.7 Current Consumption

The OTA Current Consumption is $20\mu A$.

6. LDO Integration and Performance

6.1 LDO Architecture

The designed OTA is used as the **error amplifier** in the LDO. It provides sufficient loop gain to ensure stable regulation over the output range of **0.75 V to 1.05 V**. Simulations confirm stable operation with **phase margin > 60°**, fast transient response, and **quiescent current below 60 µA**.

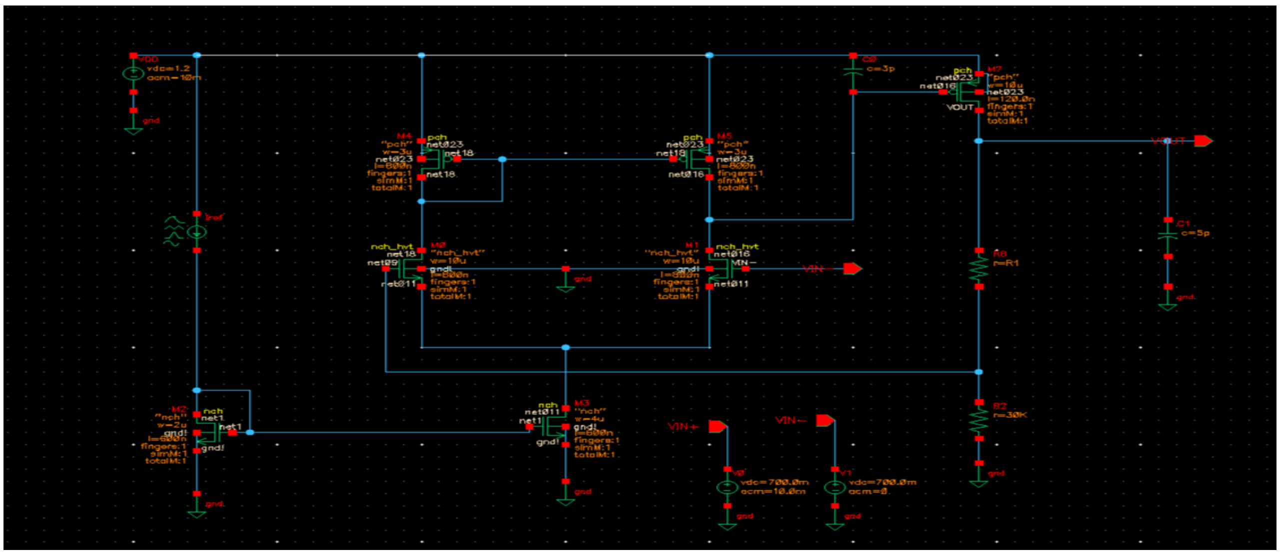


Figure 6: LDO Schematic

6.2 Programmable Resistor Network

- Divides output voltage and feeds it back to the OTA.
- R1 is digitally swept to program V_OUT in **50 mV steps**.
- R2 is fixed at 30 kΩ.

6.3 Output Voltage Equations

The general LDO output voltage is given by:

$$V_{OUT} = V_{REF}\left(1+\frac{R_1}{R_2}\right), with\, V_{REF} = 0.6\ V$$

For programmable operation, the output voltage becomes:

$$V_{OUT}(n) = V_{REF}\left(1 + \frac{R_1 + R_{prog}(n)}{R_2}\right)$$

Programming Step Size

For a desired output voltage step of **50 mV**, the required resistance increment is:

$$\Delta R_1 = \frac{\Delta V_{OUT} \cdot R_2}{V_{REF}} = \frac{0.05 \cdot 30\ \text{k}\Omega}{0.6} \approx 2.5\ \text{k}\Omega$$

R1 Sweep Range

To cover the programmable output range, $R_1$is swept from **0 to 15 kΩ**, corresponding to **8 output voltage steps**.

Programmable Output Voltage Table

Step	R_1**(kΩ)**	V_{OUT}**(V)**
0	7.50	0.75
1	10.00	0.80
2	12.50	0.85
3	15.00	0.90
4	17.50	0.95
5	20.00	1.00
6	22.50	1.05

6.4 LDO DC Regulation

DC regulation simulations show that the LDO maintains the desired output voltage across the specified range of **0.75 V to 1.05 V**. The output remains stable under nominal load conditions, confirming sufficient loop gain provided by the OTA

6.5 PMOS Pass Transistor Sizing

- PMOS: W = 10 µm, L = 120 nm
- Max load current: I_D = 30 µA

On-resistance:

$$R_{\mathrm{ON}} \approx \frac{1}{\mu_p C_{\mathrm{ox}} \frac{W}{L} (V_{\mathrm{SG}} - |\ V_{\mathrm{TH}}\ |)}$$

Assuming $\mu_p C_{\mathrm{ox}} \approx 50\ \mu A/V^2$and overdrive $V_{\mathrm{SG}} - |\ V_{\mathrm{TH}}\ | = 0.2\ V$:

$$R_{\mathrm{ON}} \approx 1.2\ k\Omega$$

Dropout voltage:

$$V_{\mathrm{DROP}} = I_{\mathrm{LOAD}} \cdot R_{\mathrm{ON}} = 23.41\mu A \cdot 1.2k\Omega \approx 28\ mV$$

m

Figure 7: LDO DC Regulation

6.3 Current Consumption

Under nominal operating conditions, the LDO draws 9.994 µA from the reference circuit, 18.59 µA from the tail current source, and 23.41 µA through the pass PMOS. The total current consumption is therefore:

$$I_{TOTAL} = 9.994 + 16.78 + 20.08 = 46.83\ \mu\text{A}$$

This total current is **below 60 µA**, confirming low-power operation while maintaining stable regulation.

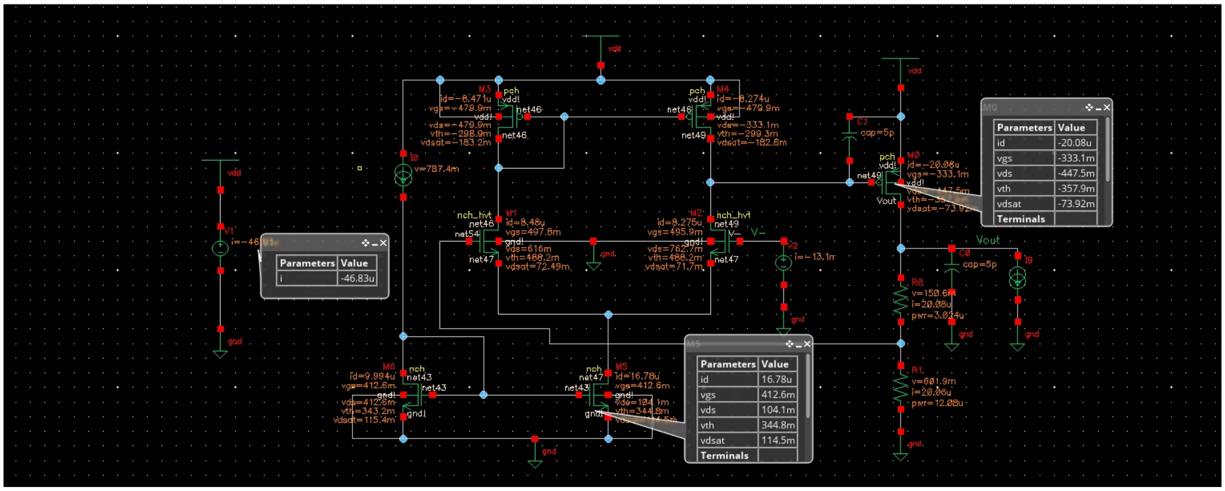


Figure 8: LDO Current Consumption

7. LDO PSRR (Power Supply Rejection Ratio)

PSRR = -58.6dB

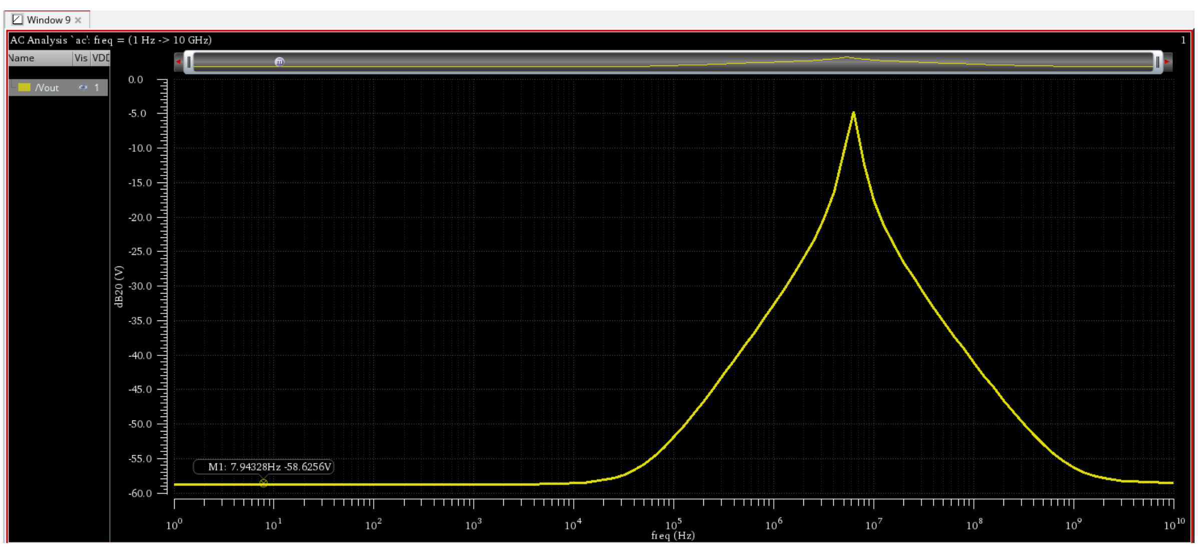


Figure 9: LDO PSRR

8. LDO PVT Analysis

PVT Analysis and Output Voltage Accuracy

A PVT analysis was performed to verify LDO robustness. The **nominal output voltage** at the typical corner is for each range is

$$752.57mV\ ,802.65mV\ ,852.92mV\ ,903.01mV\ ,953.03mV,1.003V\ ,1.053V$$

Across all PVT the worst-case deviation is **±1.5 mV (±0.2%)**, confirming stable regulation across PVT variations.

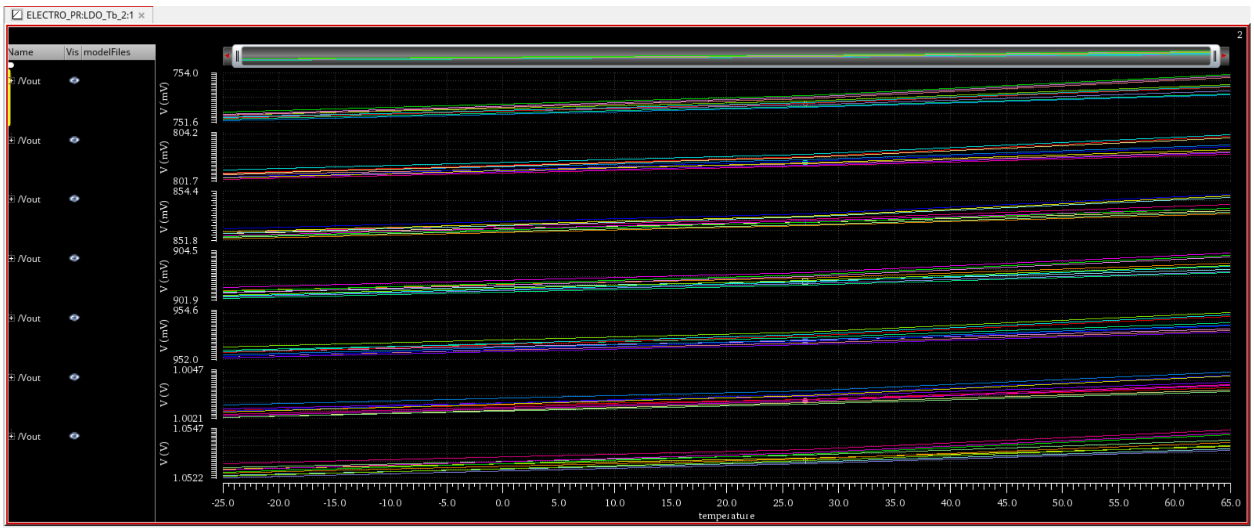


Figure 10: LDO PVT